

RESEARCH ARTICLE

A Novel Hybrid Model for Loan Default Prediction in Maritime Finance Based on Topological Data Analysis and Machine Learning

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ABSTRACT The global shipping industry, pivotal to international trade, faces inherent financial vulnerabilities due to cyclical asset values, volatile freight rates, and high leverage, rendering traditional credit risk models inadequate. This study proposes a novel framework integrating topological data analysis (TDA) and machine learning (ML) to enhance default prediction in maritime finance. By constructing correlation-based networks of shipping firms and extracting topological persistence features, such as cyclical trends and structural interdependencies, via Vietoris-Rips complexes, the model captures nonlinear risk patterns overlooked by conventional metrics. Network properties (e.g., centrality, clustering) and financial indicators (loan terms, vessel attributes, economic indices) are synthesized, enabling Graph Neural Networks (GNNs) to leverage relational and topological insights. Evaluated on a dataset of shipping loans spanning financial, operational, and macroeconomic variables, the TDA-enhanced models demonstrate significant improvements: logistic regression accuracy rose from 62.8% to 85.1%. At the same time, ROC-AUC for SVM and XGBoost reached 0.931 and 0.973, respectively. Key features like loan amount, vessel age, and debt-to-equity ratios exhibited strong correlations with defaults. Results underscore that incorporating TDA-derived persistence features improves early detection of financial distress, particularly in capturing systemic risks from interconnected market shocks. This hybrid approach offers a robust tool for lenders and policymakers to mitigate defaults in a sector critical to global commerce.

INDEX TERMS Financial risk management, loan default prediction, risk analytics, shipping industry, topological data analysis.

I. INTRODUCTION

The global shipping industry, responsible for transporting over 80% of global trade by volume, is a cornerstone of international commerce. Yet, its economic significance is matched by its vulnerability to financial instability. Historical crises—from the 2008 global financial downturn to oil price shocks and the COVID-19 pandemic—have repeatedly exposed the sector's fragility, triggering corporate bankruptcies and loan defaults [1]. This volatility stems from the industry's unique characteristics: high capital intensity, cyclical asset values, and heavy reliance on debt financing. References [2]

and [3] Shipping companies operate in an environment where freight rates fluctuate unpredictably, vessel values swing with supply-demand imbalances, and regulatory shifts—for instance, the International Maritime Organization's (IMO) 2020 sulfur cap or decarbonization mandates—introduce sudden cost pressures [4]. These dynamics render traditional credit risk models, designed for more stable industries, ill-equipped to forecast defaults in maritime finance.

The shipping sector's reliance on debt is both necessary and a liability. Acquiring vessels, which can cost hundreds of millions of dollars, demands significant upfront capital, compelling firms to leverage loans secured against volatile collateral [2]. However, the cyclicity of vessel values, driven by freight rate oscillations and geopolitical

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disruptions, creates a precarious mismatch between asset valuations and debt obligations. Compounding this risk is the industry's conservative adoption of global capital markets, which has limited access to diversified funding sources [4]. Corporate financial leverage further complicates stability, as deviations from optimal debt levels influence mergers and acquisitions (–s), deal sizes, and post-acquisition outcomes [2]. Firms operating near debt capacity thresholds face heightened vulnerability to revenue shocks, as seen during the COVID-19 pandemic, where container freight rate volatility left over-leveraged companies struggling to service loans [1].

Traditional credit risk estimation tools, such as logistic regression or discriminant analysis, rely heavily on static financial ratios (e.g., debt-to-equity, current ratios) and macroeconomic indicators. While these methods provide baseline insights, they fail to account for the shipping industry's nonlinear risk interdependencies. For example, a lender evaluating a tanker operator might overlook how a simultaneous decline in crude oil demand and tightening emissions regulations could erode freight revenues and collateral value. In this scenario, isolated financial metrics mask systemic risk. Even machine learning (ML) models, which handle high-dimensional data better than traditional tools, struggle to capture the *persistent structural patterns* underlying shipping's boom-bust cycles. Most ML frameworks prioritize local correlations, such as quarterly earnings trends, but neglect global topological features that define long-term solvency, like recurring liquidity gaps or cascading counterparty risks. This gap is critical: shipping defaults often arise not from isolated events but from the erosion of structural coherence across interconnected variables, such as charter party agreements, vessel employment rates, and bunker fuel costs.

To address these limitations, this study proposes a novel framework that integrates topological data analysis (TDA)—a mathematical approach focused on identifying shape-based patterns in data—with machine learning. TDA's strength lies in its ability to quantify *persistent homological features*, such as loops, clusters, and voids, that endure across scales of observation. In finance, these features can represent cyclical market trends, asset-liability mismatches, or latent dependencies on fragile niches. For instance, a shipping firm's financial trajectory might form a distinct topological signature in high-dimensional space, reflecting its resilience to interest rate hikes or sensitivity to freight volatility. By embedding these persistence-driven insights into ML models, lenders gain a dual advantage: (1) capturing immediate financial distress signals (e.g., liquidity ratios) and (2) identifying latent structural risks, such as unstable cash flow continuity or hidden exposure to geopolitical disruptions.

This study presents a novel predictive model tailored for the shipping industry, which applies persistence features in addition to traditional financial and industry-specific information. By enriching the feature set with outputs from TDA, the model can identify early warning signals of economic distress, even for companies with limited available

information. The application of these advanced techniques offers not only the possibility of improving prediction accuracy but also the potential for fostering a better understanding of the factors driving default risks. The paper is organized as follows: Section II reviews previous studies on loan default prediction in the shipping industry. Section III introduces the proposed model and explains how TDA features are incorporated. Section IV describes the dataset, model evaluation, and result analysis.

II. LITERATURE REVIEW

The shipping industry's reliance on debt financing and its exposure to cyclical risks have long posed challenges for lenders and investors. In [5], the authors analyze the unique financial dynamics of shipping companies, emphasizing the interplay between volatile freight rates, vessel price fluctuations, and regulatory constraints such as Basel III. They note that traditional bank financing, which historically covered 75% of shipping loans, contracted sharply after the 2008 financial crisis, forcing firms to explore capital markets through bonds, equity, and derivatives like Forward Freight Agreements (FFAs). Their study underscores the importance of integrated risk management strategies that combine operational adjustments (e.g., diversifying vessel portfolios), financial derivatives (e.g., hedging freight rate volatility), and capital structure optimization. A critical insight is that shipping companies with high leverage and spot-market exposure face heightened default risks during downturns, as seen during the 1999 and 2008 crises. They also highlight the destabilizing effects of fair-value accounting on collateral values, exacerbating liquidity constraints for banks and borrowers alike. Complementing this, [6] explicitly examines default prediction for shipping high-yield bonds, utilizing a binary logit model to identify key determinants of default risk. Their analysis highlights liquidity ratios (e.g., working capital/total assets), leverage (gearing ratio), and industry-specific factors such as freight market conditions relative to historical trends as critical predictors. Notably, their findings emphasize the heightened default risk associated with lower-rated shipping bonds, underscoring the sector's sensitivity to credit quality. The model demonstrates strong predictive performance and robustness through bootstrap validation. This study emphasizes the significance of liquidity and leverage metrics for investors, while also stressing the need for shipowners to carefully balance capital-raising efforts against potential dilution risks.

Incorporating artificial intelligence (AI) and advanced data analytics into financial risk management has emerged as a critical frontier in addressing sector-specific challenges. A study in 2018 examines the transformative potential of AI, big data analytics, and blockchain in mitigating risks across fintech applications, including peer-to-peer (P2P) lending, robo-advisory services, and cryptoassets. The study underscores systemic vulnerabilities such as credit risk underestimation in P2P platforms, compliance mismatches in automated financial advice, and fraud in decentralized crypto

markets. To address these, Giudici proposes network correlation models to capture interconnected risks and text-mining techniques to detect fraudulent activities in blockchain transactions. The paper advocates for collaborative frameworks between regulators, fintech firms, and researchers to harmonize innovation with consumer protection, emphasizing the need for “RegTech” (regulatory technology) and “SupTech” (supervisory technology) solutions [7]. This work aligns with broader calls for dynamic risk assessment tools in cyclical industries like shipping, where traditional models struggle with volatility and hidden interdependencies.

In financial risk assessment and loan default forecasting, [8] addresses the challenge of imbalanced datasets, a significant issue in maritime finance where defaults are infrequent but highly consequential. The research presents a blended methodology that integrates the Adaptive Synthetic (ADASYN) oversampling algorithm with a deep neural network (DNN) to improve class balance and enhance predictive performance. By leveraging ADASYN to generate synthetic smaller class and utilizing the DNN’s ability to capture complex nonlinear relationships in financial data, the approach demonstrates superior effectiveness to traditional models. This study highlights the limitations of conventional techniques in managing imbalanced data. It emphasizes the advantages of AI-driven methods in identifying hidden risk patterns—an essential consideration for shipping finance, where defaults often stem from interconnected systemic factors rather than isolated incidents.

A critical challenge of balancing predictive accuracy and interpretability in bankruptcy prediction models has been addressed using various methods [1]. **While interpretable**, traditional statistical techniques such as discriminant analysis and logistic regression have struggled with accuracy when dealing with a high-dimensional feature set. In contrast, advanced machine learning models, particularly tree-based methods like XGBoost and LightGBM, provide superior predictive performance but typically operate as “closed boxes,” making it difficult to understand the rationale behind individual predictions. To overcome this limitation, the authors integrate the Local Interpretable Model-Agnostic Explanations (LIME) algorithm, which generates instance-wise explanations by approximating the behavior of complex models locally with simpler, interpretable surrogate models. Their work demonstrates that the feature importance derived from LIME is generally consistent with the model’s global importance metrics, suggesting that LIME can serve as a robust tool for extracting meaningful, instance-specific insights from otherwise opaque predictive models. Furthermore, the study emphasizes the importance of instance-wise interpretation for applications such as credit rating and bankruptcy prediction, where regulatory standards demand transparency and fairness in decision-making. By comparing LIME-based explanations with the inherent feature importance provided by tree-based models, the research shows that instance-wise insights can help ensure that predictive models perform well and provide

a basis for fair treatment of loan eligibility requirements. The authors validate their approach using data on Korean companies, revealing that the local explanations maintain consistency across training and test datasets, thereby supporting the reliability of the proposed method [1].

An advanced framework has been proposed to enhance the predictive accuracy of loan default risk models by integrating ensemble machine learning techniques with robust data processing methods [9]. In their study, the authors address common challenges in credit risk prediction—including class imbalance, outlier detection, and multicollinearity—by systematically evaluating various machine learning approaches. Specifically, the study compares models such as RF, DT, SVM, AdaBoost, XGBoost, and multi-layered perceptrons, while also examining the impact of various resampling strategies such as SMOTE, ADASYN, and hybrid approaches like SMOTE combined with Edited Nearest Neighbours (SMOTE + ENNs). The experimental results demonstrate that when the dataset is properly balanced and ensemble techniques (e.g., voting and stacking) are employed, the predictive performance can reach impressive levels, as presented in the study. This comprehensive approach improves the reliability of loan default predictions and paves the way for more scalable and adaptable credit risk models in complex financial environments.

In addition, the paper emphasizes that previous research often relied on traditional statistical methods, such as logistic regression and multiple discriminant analysis, which are limited by their simplistic feature sets and inability to capture the non-linear interactions inherent in financial data. Authors argue that advanced machine learning models can overcome these limitations with effective preprocessing and feature selection techniques. Their multi-stage framework, which includes data cleaning, normalization, hyperparameter tuning, and the integration of boosting classifiers, provides a more holistic solution to the challenges of default prediction. This integrated methodology enhances model performance and offers practical insights for financial institutions looking to improve their risk assessment processes across diverse lending platforms.

A study conducted in 2018 addressed the critical need for long-term financial distress prediction in the construction industry by developing ensemble learning models capable of forecasting distress two and three years in advance [10]. Using data from 385 South Korean contractors spanning 2007–2012, financial distress was defined as two consecutive years of negative net income, avoiding reliance on legal occurrences such as bankruptcy. To handle class imbalance, they applied **SMOTE oversampling**. They tested six classifiers (SVM, ANN, C4.5, NB, LR, KNN) alongside a **voting-based ensemble** optimized via the Weighted Count of Errors and Correct Results (WCEC) metric. Their ensemble model achieved exceptional performance, outperforming all single classifiers. Notably, the ensemble combined SVM, ANN, C4.5, and KNN for two-year predictions and SVM/C4.5 for

three-year predictions. The study highlighted the construction industry's unique challenges, such as long project durations and economic sensitivity. It emphasized the practical value of early warnings for stakeholders like project owners and creditors. However, the authors acknowledged limitations, including a **region-specific dataset (South Korea)** and the exclusion of non-financial variables (e.g., management quality, macroeconomic factors), suggesting future integration of these elements for broader applicability.

Another study conducted in the field of financial distress prediction analyzed optimal models for Moroccan SMEs, comparing logistic regression (LR) and neural networks (NN) to determine their effectiveness [11]. Using data from 180 SMEs in the Fez-Meknes region, they applied LASSO and stepwise feature selection to 23 financial ratios across liquidity, solvency, profitability, and management categories. SMOTE oversampling was employed to address class imbalance. Their results highlighted **interest in sales (R14)** and **return on assets (R15)** as critical predictors, with R14 increasing distress risk and R15 mitigating it. Logistic regression models with stepwise selection demonstrated superior performance to neural networks, effectively predicting financial distress well in advance. The models exhibited strong predictive capabilities, highlighting their reliability in early risk detection, while also demonstrating Fewer Type I errors (incorrectly classifying distressed firms) compared to Type II errors. The study emphasized the robustness of logistic regression in the Moroccan context and marked the first application of LASSO for variable selection in this region. Practical implications include aiding creditors in risk assessment and providing managers with early warning signals.

The authors contextualized their work within broader literature, noting that while machine learning methods like NN often show promise, traditional statistical approaches (e.g., LR) remain competitive, particularly in region-specific settings. They contrasted their findings with studies in other regions, underscoring the importance of localized models due to varying economic conditions. The paper also addressed gaps in Moroccan-focused research, improving upon earlier works by achieving higher accuracy and integrating modern techniques like LASSO.

A novel approach to corporate bankruptcy prediction has been introduced, integrating complex network analysis with traditional predictive modeling techniques [12]. Recognizing that conventional models often rely exclusively on accounting data and financial ratios, the authors argue that critical information about a company's risk profile is also embedded in the relational ties between firms. In particular, they focus on the board of directors' networks, positing that shared directors can transmit managerial risk and influence corporate outcomes. To capture these dynamics, the study constructs several types of network structures: an undirected company–company direct network where edges represent the number of shared board members; an undirected bipartite network that links companies and directors with either weighted

or straightforward connections (the latter reflecting director dominance based on board size); and a stacked time-weighted company–company network that accounts for the decaying influence of historical board relationships over time.

The methodology proceeds in two stages. First, the authors develop new predictive drivers based on these complex network structures. By analyzing the connectivity patterns within board networks, they derive a “failure proximity score” that quantifies the risk contagion effect—essentially, the proportion of bankrupt companies within a given cluster, indicating the potential spillover of financial distress. To manage these networks' inherent high dimensionality and sparsity, Zhao et al. employ node embedding techniques, explicitly using the Node2Vec algorithm. This approach transforms the extensive, sparse network data into lower-dimensional vector representations while preserving the essential structural and semantic information. These embeddings facilitate the detection of community structures within the network and serve as input features for downstream prediction models.

In the second stage, the derived network-based drivers are integrated into a bankruptcy prediction framework and evaluated using empirical data from UK companies listed on the London Stock Exchange. The study examines various network configurations—direct, bipartite, and time-weighted networks—to assess which structure most effectively captures the inter-firm relational risk. The empirical results indicate that including these complex network-derived drivers significantly improves the predictive performance compared to models that rely solely on traditional financial ratios. Moreover, using node embeddings and community detection techniques (such as OPTICS for density-based clustering) further enhances the model's ability to capture the subtle contagion effects in corporate governance networks.

The predictive capability of industry-specific financial indicators in forecasting loan defaults for Korean shipping companies is explored, employing a logit model to develop and compare four distinct default prediction models [13]. The study evaluates traditional financial ratios—such as profitability, stability, and activity metrics—alongside shipping-specific variables, including **Freight revenue-to-sales**, **leasing expenses**, and **vessel valuation-to-total assets**. The authors find that models incorporating industry-specific indicators, particularly **operating income-to-sales**, **interest expenses-to-liabilities**, and **freight income-to-chartering cost**, significantly outperform generic models in terms of predictive accuracy. This highlights the importance of tailoring risk assessment tools to the distinct characteristics of the maritime sector, which is highly capital-intensive, subject to volatile freight rates, and strongly cyclical.

The study identifies several key financial drivers of default in the shipping sector. For instance, higher **operating income-to-sales** ratios reduce the likelihood of default, reflecting the importance of operational efficiency and revenue stability. Conversely, higher **interest**

expenses-to-liabilities and **debt ratios** increase default risk, underscoring the financial strain caused by high leverage and interest burdens in an industry heavily reliant on borrowed capital. The authors also note the paradoxical effect of the **ship value-to-total assets** ratio, where a higher concentration of ship assets increases vulnerability to market fluctuations, thereby raising default risk. This finding challenges conventional assumptions about asset concentration and stability in the shipping industry. A significant contribution of the study is its use of data from unlisted shipping companies, sourced from the Korea Shipowners' Association, which provides a more comprehensive view of the sector than studies relying solely on publicly listed firms. However, this also introduces limitations, as unlisted companies often lack the transparency and standardized reporting of their listed counterparts. Defaults in the study are defined narrowly as registration cancellations or court receiverships, which may not capture early signs of financial distress. The authors acknowledge this limitation and suggest that future research should explore broader definitions of default to improve early warning systems.

Another concern is the lack of out-of-sample validation due to the relatively short 19-year dataset. The authors recommend updating the model annually to reflect changing economic conditions and industry dynamics. Despite these limitations, the study provides critical insights for regulators, policymakers, and financial institutions, emphasizing the need to incorporate industry-specific metrics into credit risk frameworks. This is particularly relevant for Korea, where the shipping industry plays a critical role in the national economy, and systemic risks in the sector can have far-reaching economic consequences.

The determinants of default probability among internationally listed shipping corporations are investigated, employing a binary logit model on a dataset of 192 companies [14]. Unlike conventional bankruptcy prediction models focusing solely on outright defaults, this study introduces the concept of **semi-default**. In this state, firms do not go bankrupt but undergo legal status changes due to financial distress signals. This distinction allows for a more comprehensive understanding of corporate financial health in the shipping sector. The authors also address a significant gap in the literature by incorporating **financial and non-financial determinants** of default, acknowledging the unique challenges posed by the maritime industry's capital-intensive and highly cyclical nature. The study evaluates a broad set of variables affecting default probability, including **profitability indicators** (Return on Equity [ROE], Return on Assets [ROA], and Earnings Before Interest, Taxes, Depreciation, and Amortization [EBITDA]), **firm size**, and **liquidity measures** such as the current ratio. Additionally, it considers **financial leverage** as a key risk factor, given that many shipping companies rely on high levels of debt financing. Beyond firm-specific factors, the study incorporates **macroeconomic and industry-related indicators**, including freight rates and sector-specific indexes like the **Baltic Dry Index (BDI)** and

the **Dow Jones U.S. Iron & Steel Index**, proxies for broader economic and trade conditions.

The empirical findings confirm several well-established financial theories while also revealing industry-specific nuances. The results suggest that **higher profitability, larger firm size, and stronger liquidity reduce the probability of default**, reinforcing the importance of financial stability in mitigating risk. Conversely, a **high Tobin's Q**, which indicates potential market overvaluation, positively correlates with default probability, suggesting that speculative market conditions can lead to financial distress. These insights support the argument that **financial bubbles in capital-intensive industries like shipping can significantly impact credit risk**. In addition to the core regression analysis, the study provides **practical applications for financial decision-making**, particularly in **bank credit underwriting and managerial risk assessment**. By applying both **binary logit and ordered logit models**, the authors demonstrate how the effects of default predictors evolve over time and across different levels of financial distress. The study also contextualizes its findings within real-world industry events, such as the **high-profile bankruptcy of Hanjin Shipping in 2016**, which had widespread implications for global supply chains and maritime finance. This case is a stark reminder of the **systemic risks inherent in the shipping industry**, where market downturns and excessive leverage can trigger widespread financial instability.

A credit scoring model was developed to identify default risk drivers in shipping bank loans, utilizing a unique dataset of 128 loans from a ship-lending bank spanning 1997–2011 [3]. Employing a logit model with two-way clustered standard errors to address panel data biases, they highlight industry-specific factors as critical determinants of default. Key variables include freight market conditions (proxied by spreads between time-charter and spot rates), inactive tonnage levels (reflecting market oversupply), and shipowners' risk appetite (captured by time-charter policies that stabilize cash flows). Notably, financial ratios commonly used in corporate finance (e.g., leverage, liquidity) were insignificant, underscoring the shipping industry's unique volatility and reliance on market dynamics rather than static financial metrics. The study also emphasizes the role of relationship banking, with loan pricing (arrangement fees) serving as a proxy for banks' risk assessments. Their model demonstrates strong predictive power and offers practical insights for lenders, regulators, and shipowners in managing cyclical risks and loan pricing. The paper addresses gaps in maritime finance literature by integrating the 6Cs credit framework (Capacity, Capital, Collateral, Conditions, Character, Company) with shipping-specific factors, contrasting findings from general corporate loan studies. It critiques prior methodologies like UTADIS for subjectivity and advocates for empirical models tailored to shipping's capital-intensive, cyclical nature. By validating the dominance of freight market expectations and operational strategies over traditional financial metrics, the research provides a specialized tool for risk

assessment in an industry where asset values and cash flows are highly sensitive to global trade fluctuations.

The factors driving defaults in the shipping industry are empirically investigated by developing a forward-looking logit model based on a dataset of 64 shipping bonds and loans issued between 1993 and 2015 [15]. Given the capital-intensive nature of shipping, their study is motivated by the industry's sensitivity to macroeconomic fluctuations and the high stakes involved. The authors posit that a firm's financial health—captured through liquidity (working capital to total assets, WC/TA) and solvency (debt to assets, D/A)—as well as profitability changes (measured by changes in revenue-to-assets and profit margins) and macroeconomic conditions (world GDP and the VIX index), are crucial in predicting defaults. They find that higher debt levels significantly increase default risk, while better liquidity reduces it, and that macroeconomic indicators, especially the VIX, provide valuable forward-looking insights into default probabilities.

The paper sets itself apart by overcoming the limitations of previous studies that often depended on simplistic univariate analyses and were prone to overfitting. To address these challenges, the authors employ rigorous robustness checks, such as Winsorizing 1, to mitigate the influence of outliers and integrate additional control variables to stabilize the model. Adopting a logistic regression framework, they effectively model the binary nature of default events while extracting detailed marginal effects. These effects reveal the complex, non-linear relationships between the explanatory variables, such as debt ratios, liquidity measures, and macroeconomic indicators, and the probability of default. This nuanced analysis provides deeper insights into how incremental changes in these factors can significantly alter default risk. Ultimately, the study emphasizes that it is critical for banks, shipping bond investors, and credit assessment professionals to employ advanced, forward-looking credit models that account for a company's current financial health and broader economic signals when evaluating shipping defaults.

Due to persistent oversupply and a highly fragmented sector, the shipping industry has undergone substantial consolidation in recent years. Between 2008 and 2018, the sector witnessed M&A transactions totaling over \$566 billion, highlighting a marked shift toward inorganic growth strategies alongside traditional organic investments such as capital expenditures tied to the S&P and newbuilding markets. This consolidation trend reflects shipping companies' efforts to achieve scale economies, strengthen competitive positioning, and mitigate the risks associated with market volatility. Moreover, such mergers and acquisitions serve as a strategic response to intense competitive pressures and are increasingly viewed as essential vehicles for rapid expansion and operational synergies.

In this context, the reliance on debt financing becomes particularly pronounced. Shipping companies often depend on high borrowing levels to support these large-scale

transactions, which places a premium on maintaining an optimal capital structure. Excessive leverage can restrict a firm's ability to secure additional funding, hampering its strategic flexibility and growth potential. Consequently, understanding how deviations from target leverage influence M&A decisions is critical for assessing the immediate feasibility of acquisitions and evaluating their longer-term impact on firm performance and shareholder value.

Another investigation of the determinants of credit spread changes in global shipping bonds was done [4], addressing methodological gaps in prior studies. Using panel data from 54 shipping bonds across 2003–2010, they emphasize the necessity of two-way clustered standard errors to correct for biases in previous literature, which often overstated significance due to unadjusted errors. Their analysis identifies key factors: bond liquidity (proxied by market value), stock market volatility (VIX Index), cyclicalities (Merrill Lynch's GISC Index), freight earnings (via lagged freight indices), and credit ratings. Notably, liquidity and market volatility had the most substantial standardized impacts. The study contrasts pre- and post-2008 crisis periods, revealing shifts in factor significance—e.g., cyclicalities (GISC) mattered pre-crisis, while freight earnings gained prominence post-crisis. By focusing on global markets and cargo-specific issuers, they provide nuanced insights into shipping finance, though their industry-specific focus limits broader applicability. Their work underscores the importance of robust econometric methods in bond pricing research and highlights shipping's unique risks, such as freight rate volatility and cyclicalities.

A novel approach to financial time series forecasting is presented in [21], where **Topological Data Analysis (TDA)** is integrated into the N-BEATS deep learning framework. They extract three key features from persistence diagrams—**persistent entropy**, **amplitude**, and **point count**—computed over sliding windows of financial data using **Takens' 2-embedding and Vietoris-Rips complexes**. This method enables capturing non-linear structures and multi-scale temporal dynamics, often overlooked by traditional models. These TDA-derived features are inputs into the N-BEATS architecture, enhancing its forecasting performance across 32 scenarios involving cryptocurrencies and conventional financial instruments.

The empirical evaluation demonstrates that the proposed N-BEATS+TDA method consistently outperforms baseline models and other feature-enhancement strategies (e.g., lagged features, STL decomposition, time-delay embeddings) across MAPE, MAE, and RMSE metrics. The authors further confirm the statistical significance of these results through Friedman and post-hoc tests. Importantly, the study positions TDA as a classification or exploratory analysis tool and a **quantitative feature engineering method** that boosts predictive accuracy in noisy, non-stationary financial contexts. This work highlights the **forecasting utility of TDA**, filling a methodological gap in existing financial

TDA applications, which have primarily focused on market regime change detection or exploratory topology rather than forward-looking prediction.

Reference [22] proposes a novel topological data analysis (TDA) application to detect early warning signs of stock market crashes. By analyzing the S&P 500 index from 1980 onward, the authors transform financial time series into point clouds using Takens embedding, which captures multidimensional geometric structures through sliding windows. They then apply persistence homology to extract topological features (e.g., connected components, holes) from Vietoris-Rips complexes, creating persistence diagrams that reflect the stability of these features across scales. A key innovation lies in comparing successive windows using landscape distances between persistence diagrams, which reduces noise compared to traditional methods like first-derivative thresholds. The study demonstrates that TDA-based models successfully identify pre-crash volatility patterns during the 2000 dot-com bubble and 2008 financial crisis, outperforming baseline models in specificity by minimizing false positives.

While the results highlight TDA's potential as a robust tool for financial anomaly detection, the authors acknowledge limitations. The analysis is restricted to a single market (U.S. equities) and short timeframes, raising questions about generalizability across diverse markets or prolonged periods. Additionally, the method's reliance on parameter choices (e.g., embedding dimension, time delay) introduces subjectivity. Despite these caveats, the study positions TDA as a complementary approach to traditional econometrics, offering geometric insights into market dynamics that could enhance predictive frameworks.

A recent framework for univariate time series classification leverages topological data analysis (TDA), focusing on persistent homology to extract robust topological features from the data. The methodology utilizes a two-level sliding window approach, initially dividing the signal into fixed-length segments, then applying smaller, overlapping subwindows within each segment. Each subwindow undergoes time-delay embedding, after which persistence diagrams are computed. Various topological descriptors—such as persistence landscapes, Betti curves, persistent entropy, and Wasserstein distances—are derived from these diagrams. These descriptors are then aggregated (e.g., via statistical summaries like mean and standard deviation across subwindows) and used as input features for standard machine learning classifiers. Experimental evaluation on synthetic and physiological datasets, particularly in stress detection, reveals that topological features generated through this process can outperform conventional handcrafted signal features, demonstrating accuracy and generalizability across domains [23].

Study [24] proposes a distributed big data approach for internet financial fraud detection using Node2Vec graph embedding to address the limitations of rule-based systems and traditional machine learning in handling large-scale, evolving fraud patterns. By constructing financial network

graphs (nodes: entities like users, devices; edges: transactional relationships), the authors leverage Node2Vec's biased random walks (BFS/DFS) to learn low-dimensional representations of nodes, capturing structural equivalence and homophily for improved fraud pattern recognition. The framework is implemented on Apache Spark GraphX and Hadoop clusters, enabling parallel processing massive datasets. Experimental results on real-world data from a Chinese financial service provider demonstrate superior performance over DeepWalk and SVM, achieving precision rates up to 80%, recall rates near 70%, and higher F1/F2-scores, highlighting its efficacy in detecting organized fraud groups and reducing false negatives.

Despite its advancements, the study faces limitations: reliance on data from a single provider restricts generalizability across markets, and the static nature of Node2Vec limits adaptability to dynamic networks with new nodes. The authors acknowledge that incomplete network coverage could allow sophisticated fraudsters to evade detection. Future work suggests integrating inductive graph algorithms (e.g., GraphSage) to handle real-time data updates and enhance scalability in evolving financial ecosystems.

Another study in the field of financial network analysis proposes a dynamic network topology framework to examine market performance in the Chinese stock market, utilizing minimum spanning trees (MSTs) and time-varying correlation networks. Through a rolling-window approach applied to 882 stocks spanning 2006–2017, the study constructs dynamic MSTs using Pearson correlations and Kruskal's algorithm, capturing topological metrics such as normalized tree length (NTL), node centrality, and survival ratios. The results indicate that NTL drops significantly during market crises (e.g., the 2008 subprime crisis and the 2015 crash), correlating positively with returns and negatively with both volatility and tail risk. Additionally, node degree and strength distributions follow power-law behavior in 65–69% of the MSTs, emphasizing structural heterogeneity, while centrality measures identify key manufacturing-sector stocks. Edge survival ratios of 45% between consecutive windows suggest temporal stability in market structures. These findings position NTL as a potential early-warning indicator and emphasize the systemic role of the manufacturing sector [25].

III. PROPOSED MODEL

Our proposed model integrates network analysis with machine learning to address the challenge of accurate financial distress prediction. Traditional approaches often overlook interconnected corporate relationships, missing essential patterns within financial ecosystems. By constructing correlation-based networks and extracting network-centric features, our model is better equipped to detect early indicators of economic distress. This section outlines the steps of our approach.

A. STEP.1 NETWORK CONSTRUCTION

The second phase of our model focuses on constructing a correlation-based network, which plays a crucial role in understanding the intricate relationships among companies within the dataset. Initially, we compute the correlation coefficient between each pair of companies based on selected features. This results in an $N \times N$ correlation matrix, where N represents the number of companies under analysis. The correlation coefficient, denoted by r_{ij} , quantifies the linear relationship between two companies i and j , and is computed using the standard Pearson correlation formula. To transform the correlation values into a meaningful distance metric, we define the distance between two companies as $d_{ij} = 1 - |r_{ij}|$. This formula converts the correlation value into a distance, where a higher correlation results in a smaller distance, thus reflecting a stronger relationship between the companies.

Subsequently, this distance matrix is converted into a weighted network, where the weight of the edge between any two companies i and j is given by $w_{ij} = 1 - d_{ij}$. This weight reflects the strength of the company's relationship, with stronger correlations resulting in higher weights. The weighted network thus captures the underlying dependencies among companies, providing a foundation for further network-based analysis. This approach allows us to model the complex interactions often overlooked in traditional financial distress prediction models, where such interdependencies may significantly contribute to a company's risk profile.

The constructed network is the foundation for building a **Graph Neural Network (GNN)**. A GNN leverages the structural information of the network to enhance the prediction accuracy by incorporating relationships between companies, thus improving the model's ability to capture hidden patterns in the data. This novel integration of network analysis with machine learning techniques provides an innovative approach to financial distress prediction, offering deeper insights into the dynamics between companies and their financial behaviors [16], [19]. By combining these techniques, our model aims to deliver more precise and interpretable predictions than conventional methods, bridging existing gaps in the field.

B. STEP.2 TDA FEATURES EXTRACTION

In this model step, we investigate the network structure analysis by extracting **network-centric features**. These features are essential for understanding the connectivity and importance of nodes (i.e., companies) within the network, which can provide valuable insights into financial distress prediction. The key features extracted from the network include:

- a) **Degree Centrality**: This metric quantifies the number of direct connections a node has. A higher degree of centrality indicates that a company has more connections within the network, suggesting it plays a central role in the financial ecosystem [16]. In the context of economic distress, companies with a higher degree of

centrality may be more vulnerable, as they are more tightly integrated into the network and could face a larger impact in the case of a financial collapse.

- b) **Betweenness Centrality**: This measure quantifies the frequency with which a specific node lies on the shortest paths connecting other nodes. Nodes exhibiting high betweenness centrality serve as intermediaries, facilitating the flow of information between different nodes and facilitating the flow of information or resources across the network [16]. For financial distress prediction, a company with high betweenness centrality may be critical in the network, and its failure could disrupt the stability of other companies.
- c) **Closeness Centrality**: Closeness centrality assesses how close a node is to all other nodes in the network by calculating the shortest path from the node to all other nodes [16]. In financial distress prediction, companies with high closeness centrality may be better positioned to absorb shocks or quickly respond to economic crises, as they can access resources more efficiently within the network.
- d) **Clustering Coefficient**: This feature measures the tendency of a node's neighbors to be interconnected. A high clustering coefficient indicates that a company's neighbors are highly connected, forming a tightly-knit community. Companies within such clusters may be at higher risk of cascading financial distress if one company in the cluster experiences failure, as the interconnectedness facilitates the rapid spread of adverse effects [17].
- e) **PageRank Centrality**: Derived from the algorithm used by Google to rank web pages, PageRank centrality assigns more importance to nodes linked to other important nodes. This metric helps identify companies that, although they may not have the highest number of direct connections, are influential due to their connections to other central nodes. In financial distress prediction, this feature identifies companies whose failure could cause significant disruption in the network [18].
- f) **Average Neighbor Degree**: This metric evaluates a node's neighbors' average degree (or number of connections) [18]. Companies with a high average neighbor degree are likely embedded in a network of highly connected firms, making them more susceptible to external shocks or market changes.
- g) **Weighted Clustering Coefficient**: This extension of the standard clustering coefficient considers the strength or weight of the connections between nodes [19]. In financial distress prediction, this feature offers insights into the strength of company relationships, where stronger connections may indicate more robust financial interdependencies.

These extracted features are integrated into the existing financial dataset, providing a more comprehensive view of the companies' positions and vulnerabilities within the

economic network. Combining traditional financial indicators and network-based features significantly enhances the model's predictive capabilities, allowing for a deeper understanding of the factors contributing to financial distress.

Simultaneously, **community detection** is applied on the network, where a **label propagation algorithm** is used to identify clusters of companies sharing similar patterns within the network. These clusters help categorize companies based on their network behavior, providing additional categorical features incorporated into the dataset. This enriched dataset is more robust for modeling and enhances the overall accuracy of financial distress predictions.

After extracting and integrating the features, machine learning models are applied to the dataset. Various models are trained to capture patterns relevant to financial distress prediction; following this, the results are evaluated using standard performance metrics. The evaluation ensures that the predictive models effectively distinguish between economic conditions, assessing their reliability and effectiveness.

By converting correlation coefficients into distance metrics and subsequently building weighted networks, our approach effectively translates the abstract connectivity between companies into quantifiable measures of risk influence. Moreover, while prior works in TDA-based credit risk analysis concentrate on static geometrical or topological features, our model integrates static and temporal dimensions. We leverage a stacked time-weighted network structure that accounts for the persistence and decay of historical board connections over time, an aspect notably absent in the earlier studies. This temporal component is critical in the shipping industry, where long-term financial relationships and past interactions can significantly affect current loan default risk.

In addition, this methodology employs advanced node embedding techniques to extract latent, high-dimensional features from inherently sparse and complex networks. This step reduces the data's dimensionality and preserves essential relational and structural information that enhances the predictive power of the subsequent machine learning model.

The code below demonstrates an algorithmic pipeline for integrating topological data analysis (TDA) with traditional machine learning to enhance predictive modeling of shipping loan defaults. First, the dataset is loaded using pandas, and the target variable, identified by a column name containing the term "default," is separated from the feature set. The features are normalized with a StandardScaler to ensure all variables are on the same scale. Next, a Vietoris-Rips complex is constructed on the normalized data using a maximum edge length threshold, and persistent homology is calculated to identify 1-dimensional topological features, resulting in a persistence diagram. This diagram is then converted into a persistence image—a fixed-size feature representation—using a persistence image transformer with specified parameters. The persistence features are repeated and concatenated with the normalized features to form a comprehensive feature set that captures both the conventional statistical properties and the underlying topological structure of the data. The combined

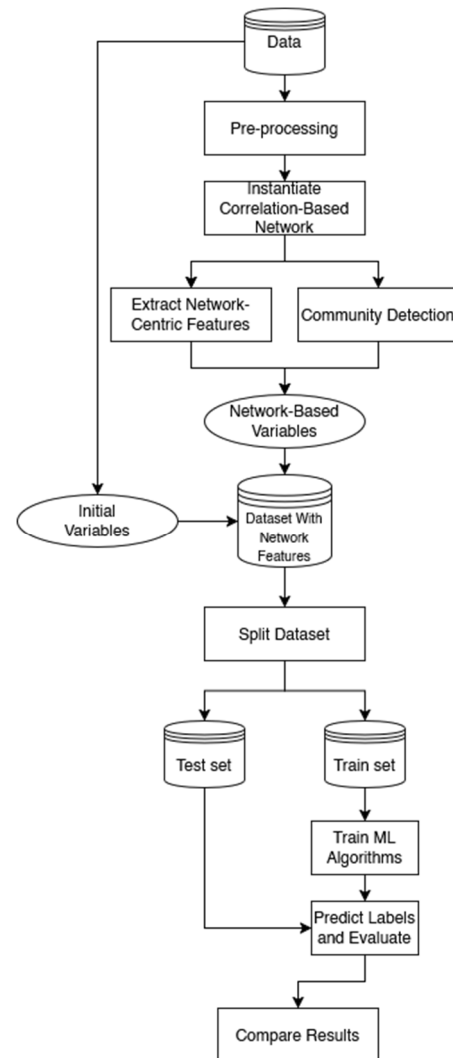


FIGURE 1. The proposed model.

features are then split into training and testing sets, and an AdaBoost classifier is defined and trained. Finally, the model's performance is evaluated by computing metrics such as accuracy and ROC AUC, demonstrating the potential benefits of incorporating TDA-derived features in the context of loan default prediction.

IV. DATA AND RESULTS

The following section presents a detailed description of the used dataset, preprocessing techniques applied, and results of the experiments performed. The dataset used in this work is composed of real-world tabular data, prepared for a binary classification task, where the target variable is a categorical label in this case, loan default status. The dataset contains a mix of numerical and categorical variables, which serve as descriptive features for each instance.

The dataset used in this work was carefully processed to ensure the quality of the input data. One of the critical steps involved **removing duplicate entries**. Duplicates, if left unchecked, can distort the model's learning process, leading

Algorithm 1 TDA Network

```

1: procedure PREDICT_DEFAULT(S, X, y, TDA, M)
2:   X_norm = Normalize(X)
3:   VR = Construct_Rips_Complex(X_norm, ε)
4:   Dgm_1 = Compute_Persistence(VR)
5:   P = Transform_Persistence(Dgm_1)
6:   X_combined = Concatenate(X_norm, P)
7:   SplitKFolds (X_train, X_test, y_train, y_test)
8:   for each model ∈ M do
9:     Train_with_HP_tuning(model, X_train, y_train)
10:    ŷ = Predict(model, X_test)
11:    Evaluate(ŷ, y_test)
12:  end if
13: end for
14: return evaluation_metrics
15: end procedure

```

to overfitting and inaccurate predictions. Removing these duplicate instances is essential because they may introduce biased patterns, reducing the model's ability to generalize well to unseen data. By eliminating such redundancies, the model is trained on a cleaner, more representative set of data, enhancing its robustness and performance.

After addressing duplicate data, the next step in the pre-processing pipeline was the standardization of features. The features were standardized using the StandardScaler from the scikit-learn package, a widely used tool for feature normalization in machine learning. The StandardScaler standardizes features by removing the mean and scaling them to unit variance, ensuring that all features are on the same scale. Specifically, the standard score for each feature is calculated using the formula:

$$z = \frac{x - u}{s} \quad (1)$$

where x represents the value of the feature, u is the mean of the feature across the training data, and s is the standard deviation of the feature. By default, the StandardScaler centers the data removing the mean and scales it using dividing by the standard deviation. This guarantees that the features are centered around zero with a variance of one. This step is essential, as it ensures that no model is biased toward features with larger numerical ranges. Features with different scales can disproportionately influence the model's learning process, especially for algorithms sensitive to the magnitude of feature values, such as Support Vector Machines. The dataset was evaluated using stratified 5-fold cross-validation, which offers a more robust and generalizable estimate of model performance by ensuring that each fold maintains the same label distribution as the original dataset. This technique partitions the data into five subsets, iteratively training on four and validating on the fifth, thus allowing every data point to be used for both training and testing. To further enhance predictive performance, hyperparameter optimization was integrated into the model training process using

GridSearchCV. This method performs an exhaustive search over specified parameter values for each model, leveraging cross-validation to identify the best combination of hyperparameters based on scoring metrics such as AUC-ROC or accuracy.

As an enhancement to the initial feature set, this study incorporates persistent homology representations extracted from the feature space using Vietoris-Rips complexes. These representations capture the underlying topological and geometric properties of the dataset, providing a complementary perspective to the conventional numerical and categorical features. By integrating persistent homology features, the objective is to leverage the capabilities of TDA¹ to improve the predictive power of machine learning models in financial distress prediction.

Persistent homology operates by identifying and analyzing features such as connected components, holes, and voids within the dataset across multiple scales. In this context, Vietoris-Rips complexes are constructed by defining simplices over data points based on their pairwise distances. As the threshold distance varies, the evolving structure of these complexes is encoded into persistence diagrams, which summarize the birth and death of topological features. This information is then transformed into numerical representations that can be seamlessly integrated into standard machine learning workflows.

The adoption of persistent homology was motivated by its potential to uncover subtle patterns and relationships within the data that might remain hidden using traditional feature engineering methods. For instance, the Vietoris-Rips approach can capture the multi-scale dependencies and interactions among variables, which are critical in understanding the dynamics of financial systems. This added layer of analysis aims to enhance the models' ability to detect early warning signals of financial distress, thus addressing a key challenge in the domain of financial prediction.

The utilized framework not only bridges the gap between TDA and traditional machine learning methodologies but also paves the way for a novel, hybrid approach that integrates geometric and statistical insights. The subsequent sections outline the step-by-step process of constructing these features and their integration into the machine learning pipeline, followed by an evaluation of their impact on predictive performance.

The primary goal of this research is to evaluate and compare the performance of standard machine learning models trained on the original feature set against those augmented with persistence-based features derived from Topological Data Analysis (TDA). By incorporating these topological features, the study investigates whether their addition enhances model performance in terms of accuracy and robustness across various experimental scenarios. This comparative analysis aims to provide insights into the potential benefits of integrating TDA into traditional machine learning workflows.

¹Topological Data Analysis.

TABLE 1. Brief description of dataset features.

Feature	Count
Amount	Loan Amount Granted to the shipping company or entity.
BDI	Baltic Dry Index – A key economic indicator that measures the cost of shipping raw materials.
Spread	A Window of Days to Deliver – The time window (number of days) allowed for the ship to deliver cargo.
Tenor	Length of Time Remaining – The remaining term or duration of the loan.
MVC	Ratio of the Value of the Ship to the Loan – A measure comparing the value of the ship to the loan granted.
ACR	Annual Cost of Repair
Balloon	Balloon Payment – A large payment due at the end of the loan term, often in shipping loan agreements.
Finance	Refers to the total Financial details regarding the loan, such as interest rate, payment schedule, etc.
DWT	Deadweight Tonnage – The total weight a ship can safely accommodate, including cargo, fuel, crew, and other essentials.
Age	Age of the Ship – The number of years since the ship was built or commissioned.
Experience	Experience of the Fleet
Fleet Size	Fleet Size of the Ship
Leverage	The debt-to-equity ratio of the shipping company, showing the proportion of debt used to finance operations.

The machine learning models employed for this evaluation include a diverse set of classifiers, namely Logistic Regression, Gradient Boosting Classifier, Random Forest Classifier, Support Vector Machine (SVM), AdaBoost Classifier, and XGBoost Classifier. These models were chosen for their widespread application and ability to represent different algorithmic families, such as linear, tree-based, ensemble, and boosting approaches. Each model was trained using both the original feature set and the augmented dataset enriched with persistence-based features.

A. DATA DESCRIPTION AND FEATURE ANALYSIS

The dataset utilized in this study pertains to loans granted to shipping companies or entities, specifically in the context of maritime shipping. It consists of a variety of features that capture different aspects of the loan and the shipping company's operations. These features provide insights into the financial health of the company, the condition and specifications of the ship, and the terms of the loan itself. The below table, data and its features are described in greater detail, alongside their respective descriptive statistics.

The features outlined in the preceding table provide a multidimensional representation of the financial, operational, and technical characteristics of the shipping companies and

TABLE 2. Statistical distribution of the dataset.

Feature	Count	Mean	Std	Skewness	Kurtosis
Amount	976	5.909581e+07	1.537133e+07	0.10	-0.18
BDI	---	3288.36	1592.47	0.03	-0.16
Spread	---	0.025	2.03	-0.14	1.45
Tenor	---	0.981	3.41	0.49	1.26
MVC	---	0.981	1.96	0.85	1.39
ACR	---	0.983	1.85	0.16	1.85
Balloon	---	0.253	1.89	-1.08	1.18
Finance	---	0.565	0.71	0.62	1.23
DWT	---	43147.11	1.84	0.03	0.01
Age	---	7.13	2.62	0.01	5.10
Experience	---	33.93	2.37	0.18	4.06
Fleet Size	---	5.10	3.98	0.31	0.48
Leverage	---	0.484	1.75	-0.45	1.49
Default ed	---	0.453	0.49	0.19	0.08

their associated loans. Each feature was meticulously selected to capture critical aspects of loan agreements, company operations, and ship-specific attributes. This comprehensive dataset facilitates an in-depth analysis of the factors contributing to loan repayment behaviors. Moreover, the inclusion of diverse variables, such as economic indicators (e.g., Baltic Dry Index), operational metrics (e.g., Deadweight Tonnage and Fleet Size), and financial measures (e.g., Leverage and Balloon Payments), ensures a holistic understanding of the dynamics influencing loan defaults. To enhance the interpretability of these features, it is essential to evaluate their statistical distributions. The subsequent table provides a summary of key descriptive statistics, including the count, mean, and standard deviation for each feature. These metrics offer insights into the central tendencies, variability, and dispersion within the dataset. Such an analysis is integral for identifying potential outliers, assessing data quality, and understanding the inherent characteristics of the variables. This step not only informs preprocessing and normalization decisions but also serves as a foundation for constructing robust predictive models in the later stages of the study.

By encompassing key dimensions such as the financial stability of borrowing entities, the operational and physical attributes of their fleets, and broader economic metrics, this dataset facilitates a comprehensive analysis of the inter-related factors influencing loan defaults. The descriptive statistics, as detailed in the subsequent table, provide a critical foundation for understanding the distribution and variability inherent in these features. This statistical characterization is instrumental in identifying underlying patterns, detecting potential anomalies, and evaluating the overall representativeness of the dataset. Furthermore, these insights are crucial in the preprocessing and feature engineering stages of the machine learning pipeline, ensuring that the models are trained on refined, balanced, and informative inputs.

The depth and heterogeneity of the dataset offer a robust basis for advancing predictive modeling efforts, paving the way for the rigorous evaluation and incorporation of sophisticated analytical methodologies in later sections of this study.

Additionally, we examine the correlation matrix for the dataset, with a particular focus on how each feature correlates with the target variable **Defaulted**. Understanding these correlations provides insight into the relationships between financial and operational characteristics and the likelihood of loan default.

- a) **Amount** shows a strong positive correlation with **Defaulted** (0.539287). This suggests that larger loan amounts are more likely to be associated with defaults. Higher loan amounts could increase the financial burden on borrowers, making it more challenging for them to meet their obligations, especially if they experience economic difficulties. This correlation is supported by research that indicates larger loans tend to have a higher risk of default.
- b) **BDI (Baltic Dry Index)** has a very weak positive correlation with **Defaulted** (0.000595). The BDI represents shipping costs and is often used as a proxy for global economic health. While this feature appears to have minimal impact on the likelihood of default in this dataset, it may indirectly reflect broader market conditions that influence financial stability. A stronger correlation might be expected in industries directly linked to global trade, such as shipping or commodities.
- c) **Spread** is negatively correlated with **Defaulted** (−0.021196). This feature likely reflects the difference in interest rates between loans or debt instruments and a benchmark, such as government bonds. A lower spread could indicate that companies with a lower cost of borrowing are less likely to default. However, the weak correlation here suggests that this variable is less influential in the model for predicting default.
- d) **Tenor**, which refers to the maturity period of a loan, shows a negative correlation with **Defaulted** (−0.213881). Longer-tenured loans may correlate with lower default rates as they provide borrowers with more time to repay. However, loans with longer tenors might also carry greater risk if market conditions worsen over time, making the borrower unable to meet longer-term obligations.
- e) **MVC (Market Value of Collateral)** shows a weak negative correlation with **Defaulted** (−0.056173). This suggests that higher collateral value may slightly reduce the likelihood of default, as collateral provides security for the lender. However, the weak correlation suggests that collateral alone is not a definitive factor in determining default risk.
- f) **ACR (Annualized Coupon Rate)** has a small positive correlation with **Defaulted** (0.044916). A higher coupon rate could indicate that a company is offering higher returns to attract investors, which might be a sign

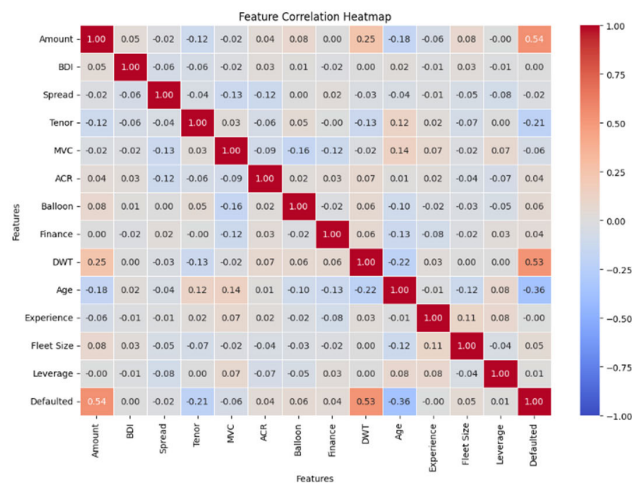
of financial distress or a higher risk profile. This could lead to a slight increase in the likelihood of default.

- g) **Balloon** (referring to balloon payments on loans) shows a positive but weak correlation with **Defaulted** (0.058492). Balloon payments, which require larger lump sums at the end of the loan term, can increase the financial strain on borrowers, thus contributing to a higher risk of default.
- h) **Finance** has a weak positive correlation with **Defaulted** (0.040964). This feature likely reflects the financing options used by a company. It could be that companies that depend more on external financing sources are more likely to face difficulties in meeting their obligations, thereby increasing their likelihood of default.
- i) **DWT (Debt Weighted Total)** demonstrates a significant positive correlation with **Defaulted** (0.527660). This suggests that a higher level of debt relative to assets significantly increases the likelihood of default. Companies with high debt burdens are generally more vulnerable to financial distress, particularly when they face liquidity issues or an inability to generate sufficient cash flow.
- j) **Age** has a moderate negative correlation with **Defaulted** (−0.359974). Older companies tend to be more established and financially stable, which may explain their lower likelihood of default. Newer companies, on the other hand, may face more challenges in managing debt and financial obligations, which increases the risk of default.
- k) **Experience** shows a very weak negative correlation with **Defaulted** (−0.004157). This indicates that the experience of a company's management or ownership has little to no impact on the likelihood of default in this dataset. However, in certain contexts, more experienced firms may have better risk management practices, thereby reducing default risk.
- l) **Fleet Size** has a weak positive correlation with **Defaulted** (0.053442). While larger fleets could indicate a company's growth and stability, they could also suggest a heavier financial burden in terms of maintenance and operational costs. The small positive correlation suggests that fleet size alone is not a strong predictor of default.
- m) **Leverage** shows a very weak positive correlation with **Defaulted** (0.013020). Leverage, which measures the amount of debt relative to equity, typically influences a company's financial health. A higher leverage ratio can indicate higher financial risk, although the minimal correlation here suggests that it is not a dominant factor in predicting default for this dataset.

In conclusion, features like **Amount**, **DWT**, and **Age** exhibit stronger correlations with **Defaulted**, suggesting that loan size, financial leverage, and the age of a company are more influential in determining default risk. On the other hand, variables like **BDI**, **Experience**, and **Leverage** show weaker

TABLE 3. Most correlated features with the target.

Feature	Correlation
Amount	0.539287
DWT	0.527660
Age	-0.359974
Tenor	-0.213881
Balloon	0.058492
MVC	-0.056173
Fleet Size	0.053442
ACR	0.044916

**FIGURE 2.** Feature correlation heat map.

correlations, indicating that their influence on default is relatively minor in this context. This correlation analysis provides useful insights into the relationships between various financial indicators and the likelihood of loan default, which can guide further feature engineering and model development.

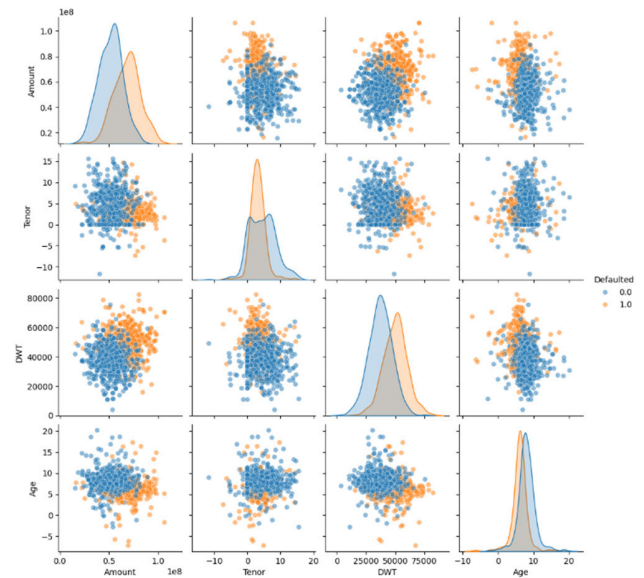
Below is a Heatmap visualization of all features correlation with the target variable.

B. RESULTS

The models' performance was assessed using various metrics, including F1-Score, Precision, Recall, ROC-AUC, and Accuracy, for both classes. To assess the impact of incorporating persistence features, each model was trained and tested under two configurations: (1) using the baseline (vanilla/plain) features and (2) using an augmented dataset with persistence features,

1) EVALUATION METRICS

To thoroughly assess model performance, several evaluation metrics are employed. These metrics provide insights into the strengths and weaknesses of a model across various aspects of classification tasks.

**FIGURE 3.** The feature pairwise chart presents a comprehensive visualization of the relationships between most correlated features, using scatterplots to depict pairwise interactions and diagonal histograms to represent individual feature distributions.

a: ACCURACY

Accuracy is a widely used metric that quantifies the proportion of correct predictions made by the model out of the total number of predictions. It is defined as the ratio of True Positives (TP) and True Negatives (TN) to the total number of instances (Equation 2). Specifically, accuracy is calculated using the following formula:

$$\text{Accuracy} = \frac{TP + TN}{\text{Total instances of data}} \quad (2)$$

This metric offers a general measure of the overall performance of the model, reflecting the correct prediction ratio across both classes, such as bankrupt and non-bankrupt companies in the context of bankruptcy prediction. Although accuracy provides an overview of the model's predictive performance, it may not be sufficient for imbalanced datasets, where one class significantly outnumbers the other. In such cases, accuracy should be considered alongside other metrics such as precision, recall, and F1 score, as they offer deeper insights into model performance [8].

b: ROC-AUC (RECEIVER OPERATING CHARACTERISTIC - AREA UNDER THE CURVE)

The Receiver Operating Characteristic (ROC) curve is a graphical tool used to visualize the performance of a binary classification model across different threshold settings. The curve plots the True Positive Rate (TPR), or recall, against the False Positive Rate (FPR) at various thresholds. The FPR is calculated as:

$$\text{FPR} = \frac{FP}{FP + TN} \quad (3)$$

The Area Under the Curve (AUC) represents the general ability of the model to discriminate between classes, with a higher AUC indicating better performance. An AUC of 1.0 signifies perfect classification, while an AUC of 0.5 suggests a model performing no better than random guessing [3].

$$AUC - ROC = \sum_{i=1}^{N-1} \frac{1}{2} (FPR_{i+1} - FPR_i) \cdot (TPR_{i+1} - TPR_i) \quad (4)$$

c: PRECISION

Precision calculates the proportion of correct positive predictions to all positive predictions made by the model. It is expressed as:

$$Precision = \frac{TP}{TP + FP} \quad (5)$$

FP in this context, stands for False Positive.

This metric assesses the model's ability to avoid false positives, measuring how often the model correctly identifies positive instances (e.g., companies at risk of bankruptcy) out of all its positive predictions. In the context of bankruptcy prediction, high precision is essential for minimizing the misclassification of financially stable companies as bankrupt, which could lead to costly or incorrect financial decisions. Thus, precision complements accuracy by offering a more targeted evaluation of the model's effectiveness in predicting positive outcomes [9].

d: RECALL

Recall measures the model's ability to identify all actual positive cases in the dataset, representing the proportion of correct positive predictions among all actual positives. It is calculated as:

$$Recall = \frac{TP}{TP + FN} \quad (6)$$

This metric reflects how well the model avoids false negatives, meaning it quantifies how many of the actual bankrupt companies (true positives) are successfully identified by the model. In scenarios like bankruptcy prediction, where the cost of missing a true positive (i.e., failing to identify a bankrupt company) is high, recall is crucial. A higher recall ensures fewer false negatives and suggests that the model captures a larger proportion of distressed companies. Recall, when considered alongside precision, provides a more comprehensive understanding of the model's ability to correctly identify bankrupt companies [9].

e: F1-SCORE

The F1-Score is the harmonic mean of Precision and Recall, offering a balanced assessment that considers both false positives and false negatives. It is especially valuable in situations with class imbalance, as it ensures the model is not evaluated based on just one metric. The F1-Score is valued for offering a single value that captures both the accuracy and

the completeness of the model's positive class predictions. A higher F1-Score indicates a better balance between precision and recall, reflecting the model's overall performance in classifying the positive class [20].

$$F1Score = \frac{2 \times (Precision \times Recall)}{Precision + Recall} \quad (7)$$

2) CLASSIFICATION MODELS

Models utilized in this study are as following:

a: LOGISTIC REGRESSION

Logistic Regression demonstrates notable improvements when augmented with persistence features, showing enhanced accuracy and a stronger capacity to differentiate between default and non-default cases. In particular, both precision and recall for the default class improve, underscoring the model's increased ability to correctly identify high-risk instances. This leads to a more balanced performance, reflected in higher F1-scores for the default class, making the model more reliable in predicting loan defaults within the shipping industry.

b: RANDOM FOREST CLASSIFIER

The Random Forest Classifier shows robust results in both baseline and enhanced feature settings, demonstrating further improvements when persistence features are incorporated. While the model already performed well with the original features, the addition of persistence information leads to increased sensitivity and precision, as evidenced by higher metrics for the default class. This highlights the classifier's augmented ability to correctly identify defaulting companies, thereby strengthening its overall reliability in predicting loan defaults within the shipping industry.

c: SUPPORT VECTOR MACHINE (SVM)

The incorporation of persistence features led to significant enhancements for the SVM model. The model's overall ability to distinguish between default and non-default cases improved, with a marked increase in its sensitivity to identifying positive instances. This improved balance between false positives and false negatives underscores the enhanced discriminatory power of the model, making it more effective for predicting loan defaults in the shipping industry.

d: GRADIENT BOOSTING CLASSIFIER

Gradient Boosting exhibited consistent enhancements across various performance measures with the addition of persistence features. The improvements in overall accuracy and discriminatory power, alongside better identification of positive instances, indicate a more reliable model for predicting loan defaults. Enhanced recall and precision for the default class contribute to a more balanced performance, strengthening the model's effectiveness in recognizing at-risk companies in the shipping industry.

TABLE 4. Vanilla models performance across different metrics.

metric	LR Vanilla	RF Vanilla	SVM Vanilla	GB Vanilla	AdaBoost Vanilla	XGBoost Vanilla
Accuracy	0.681	0.843	0.776	0.846	0.796	0.843
ROC	0.683	0.897	0.809	0.905	0.861	0.899
Precision (Class 0)	0.71	0.86	0.74	0.88	0.80	0.88
Precision (Class 1)	0.58	0.85	0.81	0.84	0.72	0.85
Recall (Class 0)	0.81	0.92	0.94	0.91	0.85	0.92
Recall (Class 1)	0.45	0.76	0.46	0.80	0.65	0.78
F1-Score (Class 0)	0.75	0.89	0.83	0.89	0.82	0.90
F1-Score (Class 1)	0.50	0.80	0.59	0.81	0.68	0.81

e: ADABOOST CLASSIFIER

The AdaBoost classifier showed considerable enhancements with the integration of persistence features. Its improved overall performance, along with marked gains in both precision and recall for the positive class, indicates a stronger capability to balance false positives and false negatives. These improvements make the classifier more reliable for identifying at-risk companies in the shipping industry.

f: XGBOOST CLASSIFIER

XGBoost maintained robust performance across both configurations, with the addition of persistence features yielding consistent improvements. The model’s overall predictive ability and its effectiveness in identifying default cases were further enhanced, as reflected in improved metrics for precision, recall, and F1-Score for the positive class. These enhancements underscore the model’s solid performance in forecasting loan defaults within the shipping industry.

The integration of persistence features into the machine learning models significantly improves their overall performance across multiple evaluation metrics, demonstrating the potential of topological data analysis for enhancing predictive modeling in this domain. Each model exhibits notable improvements in their performance after the inclusion of persistence features, with Logistic Regression (LR) showing the most pronounced gains. Specifically, LR experiences a substantial increase in accuracy.

The table 4 presents the performance metrics for Vanilla models. It is observed that XGBoost outperforms other models across various evaluation criteria.

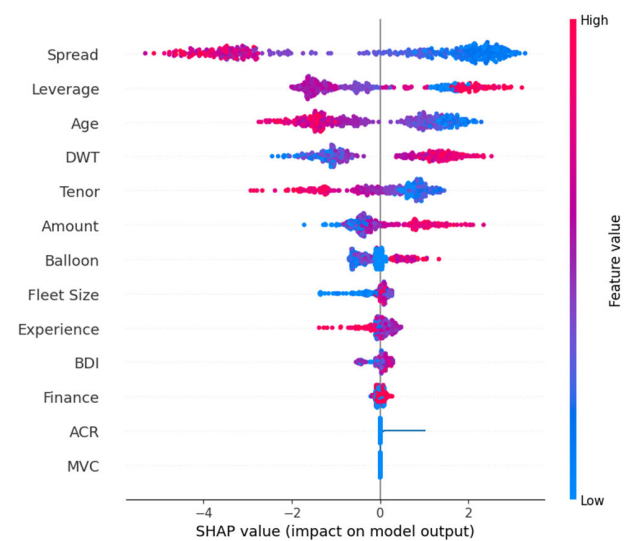


FIGURE 4. SHAP chart for XGBoostClassifier.

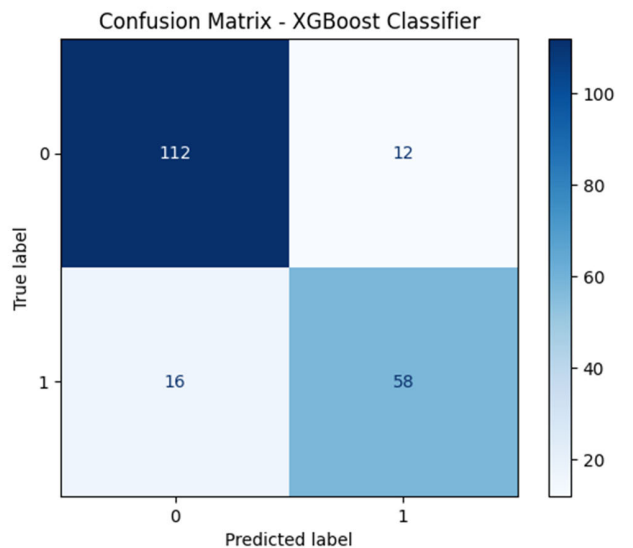


FIGURE 5. Confusion Matrix for XGBoostClassifier.

The performance metrics for other models, such as Gradient Boosting (GB), AdaBoost, and XGBoost, also reflect positive improvements in accuracy, precision, recall, and F1-score after incorporating persistence features. These models exhibit an increase in both their ability to predict Class 0 and Class 1 instances, with F1-scores for both classes rising across all models. The consistent performance gains across all the models suggest that persistence features provide valuable additional information that enhances their predictive power.

The SHAP (SHapley Additive exPlanations) summary plot (Figure 4) provides a robust visual representation of the contributions of individual features in the XGBoostClassifier model for predicting loan default. The vertical axis lists the features sorted in descending order of their average absolute SHAP values, reflecting each variable’s overall influence on the model’s output. Notably, features such as *Leverage*,

Amount, and *Age* manifest a wide distribution of SHAP values, underscoring their critical role in shaping the prediction landscape.

On the horizontal axis, the SHAP values are delineated such that a positive value drives the prediction towards a default outcome, whereas a negative value favors a prediction of no default. The dispersion of points across this axis for each feature indicates the degree of variability in its impact across different instances. Specifically, a more extensive spread denotes a feature’s capacity to significantly alter the prediction in either direction, suggesting that the influence of these features is not uniform across the dataset.

The color gradient of the points, ranging from blue (indicative of lower feature values) to red (denoting higher feature values), affords an additional layer of interpretability. This visual encoding permits the identification of non-linear effects and potential thresholds; for instance, red points consistently occupying the positive side of the SHAP axis for *Leverage* imply that high leverage levels markedly increase the likelihood of default. Conversely, the presence of blue points on the negative side reinforces that lower levels of the same feature favor stability and reduced default risk. Beyond individual features, the aggregated insights gleaned from the SHAP summary plot substantiate the model’s transparent and explainable nature. By correlating feature values with their corresponding impact on the model’s predictions, stakeholders are provided with a granular understanding of how specific financial and operational variables—such as loan amount, asset age, and maritime operational metrics—collectively drive risk assessment outcomes. This is particularly relevant in the shipping industry context, where high leverage, substantial loan amounts, and older assets can be emblematic of increased financial vulnerability due to elevated maintenance costs or market competitiveness challenges.

In summary, the SHAP summary plot not only corroborates the model’s predictive reliability through the clear delineation of influential features, but also contributes to enhanced interpretability by elucidating the directional impact of feature values on prediction outcomes. This dual functionality of performance evaluation and interpretability strengthens the model’s utility as a transparent decision-support tool in loan default risk assessment within the shipping industry.

Additionally the confusion matrix on the right evaluates the classifier’s predictive accuracy by displaying the agreement between the true and predicted labels. The matrix reveals that the model correctly identified 112 instances of “No Default” and 58 instances of “Default,” corresponding to true negatives and true positives, respectively. However, it also indicates 12 false positives and 16 false negatives. These results demonstrate the classifier’s strong predictive performance overall. Together, the SHAP chart and confusion matrix provide both feature-level explainability and a clear assessment of model performance, aligning with the goal of achieving interpretable and reliable results in classification tasks.

TABLE 5. Persistence models performance across different metrics.

metri c	LR Persis tence	GB Persis tence	RF Persis tence	AdaB oost Persis tence	SVM Persis tence	XGBo oost Persis tence
Accu racy	0.862	0.938	0.936	0.946	0.863	0.937
ROC	0.920	0.980	0.973	0.967	0.931	0.973
Preci sion Class 0	0.85	0.89	0.89	0.91	0.84	0.89
Preci sion Class 1	0.82	0.93	0.93	0.94	0.89	0.92
Recal 1 Class 0	0.92	0.94	0.98	0.95	0.91	0.93
Recal 1 Class 1	0.76	0.86	0.85	0.89	0.81	0.93
F1- Score Class 0	0.86	0.92	0.92	0.93	0.88	0.93
F1- Scor e Class 1	0.81	0.89	0.89	0.91	0.85	0.91

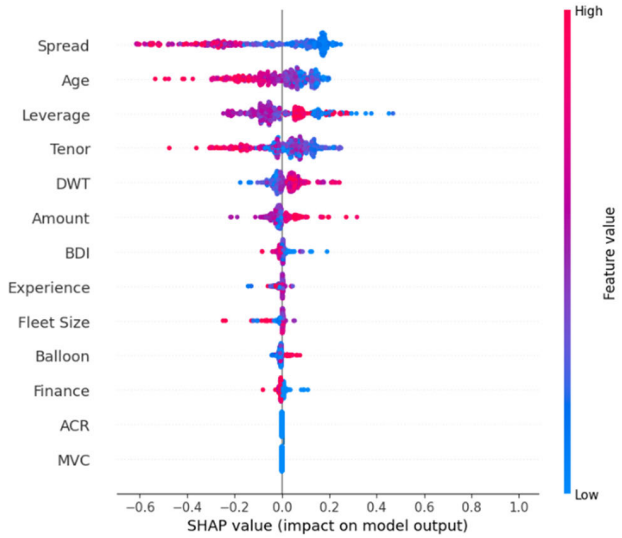


FIGURE 6. SHAP chart for AdaBoostClassifier.

The table 5 summarizes the performance of models enhanced with persistence features, comparing the results of the vanilla models it is observed that persistence model mostly outperform their vanilla counterparts.

The SHAP (SHapley Additive exPlanations) summary plot (Figure 6) offers a comprehensive depiction of each feature’s influence on the AdaBoostClassifier’s prediction of

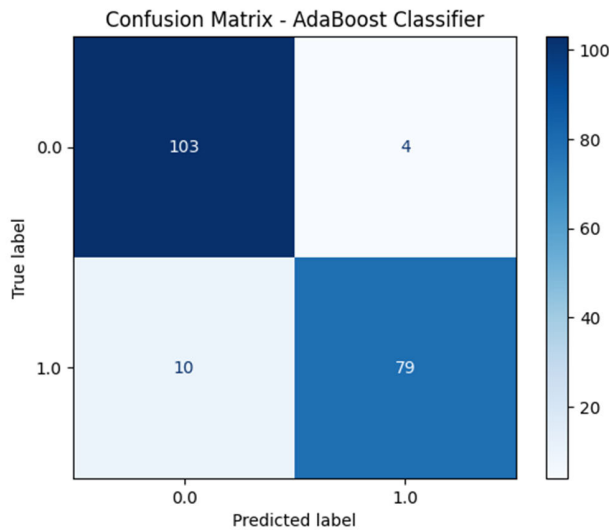


FIGURE 7. Confusion Matrix for AdaBoostClassifier.

loan default. By ordering the features according to their average absolute SHAP values, this visualization underscores the relative impact of each variable within the model's decision framework. Notably, variables such as *Spread*, *Age*, and *Leverage* demonstrate particularly extensive variability in their contributions, thereby highlighting their central role in the classifier's output determination.

In addition to quantifying feature importance, the plot employs a color gradient to distinguish between high and low feature values, where red signifies elevated values and blue indicates lower ones. This chromatic encoding facilitates a nuanced interpretation of the model's behavior, as it reveals how certain feature magnitudes are associated with an increased tendency toward default, while lower values correlate with a reduced risk. Such insights enable a deeper understanding of the underlying dynamics that drive the AdaBoostClassifier's predictive performance.

Moreover, the dispersion of points for each feature underscores the heterogeneity of its effects across different observations in the dataset. For example, while *Spread* and *Leverage* clearly manifest broad distributions of SHAP values, features such as *Tenor* or *MVC* display relatively narrower spreads, indicating more stable or modest effects on the output. This nuanced view of feature interactions and individual contributions allows for a granular assessment of how various economic, operational, and contractual parameters jointly affect default risk in maritime finance.

In parallel, the confusion matrix corroborates the robustness of the AdaBoost model by juxtaposing predicted labels against actual outcomes. The model's strong capability to correctly identify 103 instances of "No Default" and 79 instances of "Default," coupled with minimal misclassifications (four "No Default" instances predicted as "Default," and ten "Default" instances predicted as "No

Default"), underscores its balanced performance across classes. By accurately capturing the risk of default while maintaining low false-positive rates, the model is well suited for critical financial decision-making in the shipping industry context.

The marked improvements observed in accuracy, ROC AUC, and precision-recall metrics upon incorporating persistence features affirm that these variables capture salient temporal or sequential patterns fundamental to loan default prediction. Specifically, the AdaBoost model displays a pronounced rise in accuracy (0.796 to 0.94) and ROC AUC (0.860 to 0.967), which is matched by similar gains in precision, recall, and F1-scores for both classes. These findings substantiate the hypothesis that persistence features bolster the ensemble's capacity to discriminate between resilient and at-risk loans. Consequently, their inclusion not only enhances the interpretability of model outputs—through SHAP's elucidation of critical risk drivers—but also augments the model's overall predictive performance, demonstrating their utility in robust credit risk assessment.

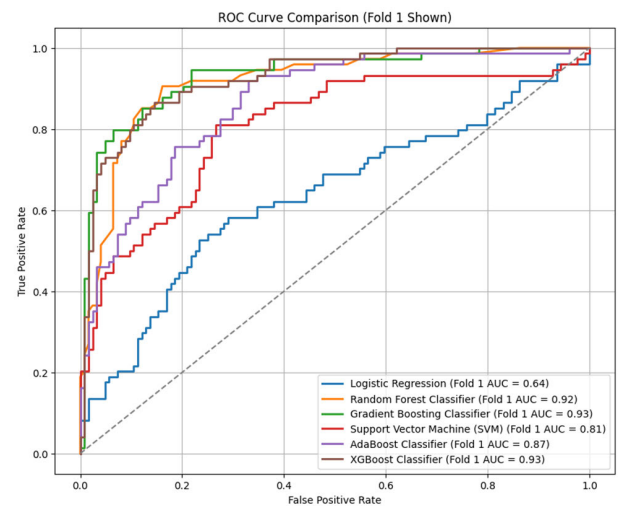


FIGURE 8. Vanilla models performance across different metrics.

Figure 5 presents the ROC curves for various classification models trained without incorporating persistence features, enabling a comparative evaluation of their performance. The models' effectiveness is quantified by the area under the ROC curve (AUC), with higher values reflecting greater discriminatory power. The XGBoost Classifier achieves the highest AUC of 0.93, closely followed by the Gradient Boosting Classifier with an AUC of 0.93. These results suggest that ensemble methods, particularly those employing boosting techniques, exhibit outstanding performance in this context. The Random Forest Classifier and AdaBoost Classifier also demonstrate robust performance, with AUC values of 0.92 and 0.87, respectively. In contrast, the Support Vector Machine (SVM) and Logistic Regression models yield comparatively lower AUC values of 0.83 and 0.64, respectively, indicating relatively poorer classification performance. These

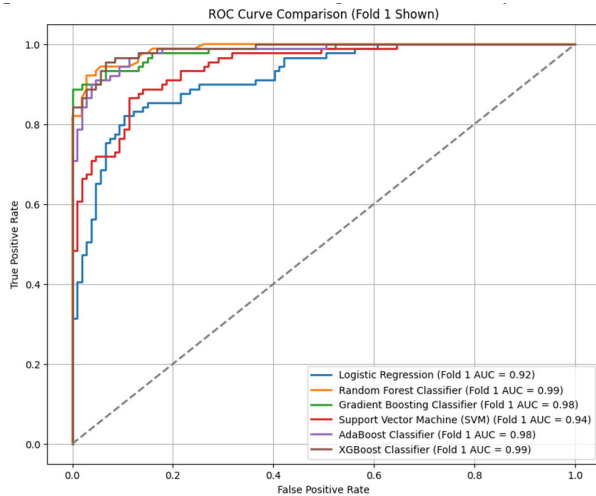


FIGURE 9. Persistence enhanced models performance across different metrics.

findings underscore the efficacy of ensemble methods and the potential importance of persistence features in enhancing model accuracy.

Figure 6 presents the ROC curve comparison for various classification models trained on a dataset enhanced with persistence features. The models exhibit notably improved performance, as evidenced by their elevated AUC values. Specifically, Logistic Regression achieves an AUC of 0.92, Random Forest Classifier attains 0.99, Gradient Boosting Classifier (presumably a correction for “Conducts Boosting”) reaches 0.98, Support Vector Machine (SVM) scores 0.94, and AdaBoost Classifier achieves 0.98. These results signify a significant enhancement in classification performance across all models compared to their previous counterparts without persistence features. Notably, Logistic Regression shows the most substantial improvement, jumping from an AUC of 0.64 to 0.92, highlighting the substantial contribution of persistence features in capturing critical patterns within the data.

The boosting algorithms, particularly Gradient Boosting and AdaBoost, continue to outperform other models, maintaining their superior performance even with the inclusion of persistence features. This suggests that these models are particularly adept at leveraging the additional information provided by persistence features. The overall improvement in performance across all models underscores the importance of feature engineering, particularly the integration of persistence features, in enhancing the discriminatory power of classification models. These findings have implications for model selection and suggest that incorporating persistence features could be a valuable strategy in improving classification accuracy in similar datasets.

3) MANAGERIAL INSIGHTS BASED ON THE SHAP AND CORRELATION MAP FOR LOAN GRANTING DECISIONS

The SHAP analysis and correlation heatmap collectively provide significant insights for managerial decision-making

regarding loan allocation to shipping companies. The SHAP analysis highlights key determinants such as **Leverage**, **Age**, **DWT**, and **Spread**, with high leverage emerging as a strong indicator of increased default risk. This finding underscores the importance of prioritizing loans to companies with stable financial metrics and moderate leverage levels to minimize the likelihood of default. Furthermore, operational characteristics like **Age** demonstrate that mature and experienced firms are less risky, suggesting that such companies are better candidates for loan approvals.

The feature correlation heatmap reinforces these insights by illustrating relationships among variables. A positive correlation between **Leverage** and default risk and a negative correlation between **Age** and default risk indicate that younger companies or those with high leverage require stricter evaluation protocols. Conversely, firms exhibiting operational maturity and moderate loan sizes present safer opportunities. These findings highlight the necessity of integrating financial and operational metrics into a comprehensive risk assessment framework to ensure robust credit evaluations.

Based on these results, managerial strategies should focus on a risk-based loan allocation framework. Companies with **low leverage**, **moderate fleet sizes**, and **favorable operational metrics** can be prioritized for standard loan products, while high-risk companies may require alternative solutions such as collateral-backed loans or performance-based repayment structures. A data-driven approach leveraging SHAP and correlation insights can significantly enhance credit assessment processes, ensuring sustainable financial outcomes for both the bank and its clients while supporting the growth of reliable firms in the shipping sector.

a: LOW-RISK COMPANIES

For companies characterized by **low leverage**, **moderate fleet sizes**, and **favorable operational metrics**, managerial strategies should prioritize offering standard loan products with competitive terms. These firms demonstrate financial stability and lower default probabilities, making them ideal candidates for less restrictive financing. Banks can consider extending credit lines or unsecured loans to such companies, enabling them to expand operations or invest in efficiency-enhancing technologies. Additionally, proactive engagement, such as regular financial reviews and operational feedback, can help these companies maintain their favorable risk profiles while strengthening client relationships.

b: HIGH-RISK COMPANIES

High-risk companies—those with **high leverage ratios**, **small or excessively large fleet sizes**, and **weaker operational metrics**—require a different managerial approach to mitigate potential financial losses. For these companies, banks should adopt collateral-backed loan products or performance-based repayment structures. By linking repayment terms to predefined performance metrics, banks

can incentivize better financial and operational management. Furthermore, introducing credit enhancements, such as guarantees or insurance products, can reduce exposure to defaults while providing these firms with necessary funding. Comprehensive risk assessments and close monitoring are crucial for maintaining control over credit exposure.

c: MODERATE-RISK COMPANIES

Companies with metrics that fall between low-risk and high-risk categories represent a balanced opportunity. For these firms, managerial decisions should focus on tailored financial products that combine the benefits of standard loans with additional risk mitigants. For instance, offering loans with adjustable interest rates based on performance can encourage improved financial discipline. Banks should also provide advisory services to help these companies optimize operations and improve financial health. Building long-term partnerships with moderate-risk firms can yield significant returns as these companies strengthen their positions and transition toward lower-risk profiles.

V. CONCLUSION

The global shipping industry's inherent financial volatility necessitates advanced risk assessment tools capable of capturing nonlinear interdependencies and systemic risks. This study addresses this challenge by introducing a novel framework that integrates topological data analysis (TDA) with machine learning (ML) to enhance default prediction in maritime finance. By constructing correlation-based networks of shipping firms and extracting persistent homological features—such as cyclical trends and structural dependencies—via Vietoris-Rips complexes, the model effectively identifies latent risk patterns overlooked by conventional metrics. Network properties (e.g., centrality, clustering) and financial indicators are synthesized, enabling Graph Neural Networks (GNNs) to leverage both relational and topological insights.

Evaluated on a comprehensive dataset of shipping loans that encompassed financial, operational, and macroeconomic variables, the TDA-enhanced models demonstrated substantial performance improvements. The integration of topological persistence features enabled the models to better capture systemic risks and complex interdependencies within the shipping industry. Key predictors, such as loan amount, vessel age, and debt-to-equity ratios, were identified as significant factors influencing default risk, aligning with the unique dynamics of the maritime sector. SHAP analysis further reinforced the interpretability of these features, emphasizing their critical contributions to the decision-making process of the model. These findings highlight the effectiveness of combining TDA with machine learning to enhance predictive accuracy and provide deeper insights into financial risk assessment.

The findings have practical implications for lenders and policymakers. By prioritizing firms with stable leverage

ratios, mature fleets, and favorable operational metrics, financial institutions can mitigate default risks. High-risk entities may benefit from collateral-backed loans or performance-linked repayment structures. This hybrid TDA-ML approach not only enhances predictive accuracy but also provides actionable insights for credit assessment, fostering resilience in a sector vital to global trade.

Limitations include reliance on region-specific data and the exclusion of non-financial variables like geopolitical events. Future work could expand the dataset to include global shipping firms, integrate real-time macroeconomic indicators, and explore dynamic network representations to capture evolving market conditions. Extending this framework to other cyclical industries, such as aviation or energy, could further validate its generalizability. Overall, this research advances the frontier of financial risk analytics, offering a robust, interpretable tool for navigating the complexities of maritime finance.

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